

Problem

Given a string S, find the smallest positive length P such that the string can be formed by repeating its prefix of length P

Input Format

A single line containing the string S.

Constraints

$1 \leq |S| \leq 10^5$ S contains lowercase English letters.

Submissions

Output Format

Print the smallest period of the string.

Sample Input 0

ababab

Sample Output 0

2

Leaderboard

Discussions

[Change Theme](#)

Language

Java 15



```

1  import java.io.*;
2  import java.util.Scanner;
3  public class Solution {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          if(!sc.hasNext()) return;
7          String s = sc.next();
8          char[] str = s.toCharArray();
9          int n = str.length;
10         int[] arr1 = new int[n];
11         int j=0;
12         for(int i=1;i<n;i++) {
13             while(j>0 && str[i]!=str[j]) {
14                 j=arr1[j-1];
15             }
16             if(str[i]==str[j]) {
17                 j++;
18             }
19             arr1[i]=j;
20         }
21         int border = arr1[n-1];
22         int period = n-border;
23         if(n % period==0) {
24             System.out.println(period);
25         }else {
26             System.out.println(n);
27         }
28     }

```

Line: 29 Col: 2